



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.DS.1

Statistical Questions and Data

Flagship

Lesson 8-1-flagship · Teacher Copy

Name: _____

Date: _____

Class: _____

FRONT OFFICE MISSION

Scout the Hardwood

You are the new data analyst for the Riverside Hoops youth league. The head coach drops a stack of questions on your desk — some will yield rich data full of variety, others just one fixed fact. Master statistical questions and you decide what the front office actually investigates.



TODAY'S OBJECTIVES

Content Objective: I can tell the difference between a statistical question and a non-statistical question.

Language Objective: I can explain my reasoning using the words statistical question, data, variability, and survey.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Statistical Question

Data

Variability

Survey

Data distribution



Tap any word to see what it means and a picture.

1

A question that expects a variety of answers and is answered by collecting data is a

2

Facts or numbers collected to answer a question are .

3

How much the values in a data set differ from one another is the

4

A way to collect data by asking many people the same question is a

5

The way data values are spread out across a range is the .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

While sorting the questions, what clue tells you instantly that a question will need DATA from many players to answer?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Statistical Question** — A question where the answers will be different for different people.

Pregunta estadística — Una pregunta cuyas respuestas serán distintas para distintas personas.

- ☐ **Data** — Facts and numbers you collect, like answers to a survey.

Datos — Datos y números que recoges, como respuestas a una encuesta.

- ☐ **Variability** — How spread out the numbers are.

Variabilidad — Qué tan separados están los números.

- ☐ **Survey** — Collecting facts by asking people questions.

Encuesta — Recoger datos haciendo preguntas a las personas.

- ☐ **Data distribution** — How the data looks: where it sits and how spread out it is.

Distribución de datos — Cómo se ven los datos: dónde están y qué tan separados están.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I know it needs data because it asks about ____.

Sé que necesita datos porque pregunta sobre ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A statistical question expects answers that vary across many cases.

Evidence: Students identify "How tall is each player?" as statistical because heights vary, while "How tall is our point guard?" has one fixed answer.

Reasoning: This evidence connects the definition of statistical question to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I know it needs data because it asks about ____.**

Sé que necesita datos porque pregunta sobre ____.

- **The answers will be different because ____.**

Las respuestas serán diferentes porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Statistical Question
2. **Notes 2:** Data
3. **Notes 3:** Variability
4. **Notes 4:** Survey
5. **Notes 5:** Data distribution
6. **Watch for this mistake:** *A common mistake in Statistical Questions and Data is judging a question by how many people or things it names, instead of checking whether the quantity being asked about actually varies. Students see a question about ONE player, ONE team, or ONE season — like "How many touchdowns did the quarterback throw this season?" — and assume it must be non-statistical since only one person is named. But a season includes many games, so the number of touchdowns changes from game to game, which makes it statistical. Just as often, students see the word "each" attached to a group and assume every group question is automatically statistical — but "How many players are on the team?" asks about the whole roster, and every player would report the exact same number, so it is NOT statistical. Before you decide, ask: if I actually collected this data, would every answer come back the same, or would it vary?*
7. **Try It 1:** A. How many rebounds does each player grab per game? — *"How many rebounds does each player grab per game?" is statistical because different players grab different numbers of rebounds — the data anticipates variability. The other three each have one fixed answer.*
8. **Try It 2:** A. Because different players warm up for different amounts of time — *A statistical question anticipates variability. Different players spend different amounts of time warming up, so the collected data will vary from player to player.*